



**COLLEGE OF ENGINEERING
DEPARTMENT OF SOFTWARE ENGINEERING**

Software Requirement Specification

Author: Abdulkerim Adem

Submit Date: 1/10/2026

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Software Requirements Specification (SRS)

Project Name: EduVox
Platform: Mobile Application (Android)
Version: 1.0
Date: January 10, 2026

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a detailed description of the functional and non-functional requirements of **EduVox**, a mobile e-learning application designed for the Ethiopian market.

The document serves as a reference for developers, designers, testers, and stakeholders to understand system behavior, constraints, and expectations.

1.2 Scope

EduVox is a Flutter-based mobile application that connects **Instructors** and **Students** through an online learning platform with localized payment integration.

The system enables instructors to create and manage educational content, while students can browse, purchase, and consume courses.

Key features include:

- Course creation and management by instructors
- Course enrollment and progress tracking for students
- Secure authentication and role-based access
- Integrated Ethiopian payment gateway (Chapa)
- Rating and review system
- User profile and theme customization

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
SRS	Software Requirements Specification

UI	User Interface
UX	User Experience
MVP	Minimum Viable Product
ETB	Ethiopian Birr
UID	Unique Identifier
API	Application Programming Interface

1.4 References

- IEEE 830-1998 Software Requirements Specification Standard
- Flutter Documentation
- Firebase Documentation
- Chapa Payment API Documentation

1.5 Overview

This document is organized as follows:

- Section 2 describes the overall system
- Section 3 outlines specific system requirements
- Section 4 details external interfaces
- Section 5 covers non-functional requirements
- Section 6 lists future enhancements

2.Overall Description

2.1 Product Perspective

EduVox is a standalone mobile application developed using Flutter and Firebase. It uses Firebase Authentication, Firestore, and Storage services, with Chapa integrated for payments.

2.2 Product Functions

- User authentication and role management
- Course creation, publishing, and enrollment
- Lesson content delivery (video and documents)

- Payment processing and verification
- Course progress tracking
- Course rating and analytics

2.3 User Classes and Characteristics

2.3.1 Student

- Can browse and search for courses
- Can enroll in free or paid courses
- Can consume video and document lessons
- Can track learning progress and rate courses

2.3.2 Instructor

- Can create, update, publish, and delete courses
- Can upload lessons and manage content
- Can view student enrollments and revenue analytics

2.4 Operating Environment

- Mobile Devices: Android & iOS
- Framework: Flutter (Dart)
- Backend: Firebase (Firestore, Auth, Storage)
- Payment Gateway: Chapa
- Internet connectivity required for content access

2.5 Design and Implementation Constraints

- Payment integration depends on Chapa API availability
- Firebase Firestore NoSQL structure must be used
- Chapa secret key must be protected in production

2.6 Assumptions and Dependencies

- Users have internet access

- Chapa supports all required Ethiopian payment methods
- Firebase services remain operational

3. Specific Requirements

3.1 Functional Requirements

3.1.1 Authentication and Authorization

- **FR-01:** The system shall allow users to register using Email/Password or Google Sign-In.
- **FR-02:** The system shall require users to select a role (Student or Instructor) during registration.
- **FR-03:** User roles shall be immutable after account creation.
- **FR-04:** The system shall authenticate users and redirect them to role-specific dashboards.
- **FR-05:** The system shall persist login sessions across application restarts.
- **FR-06:** The system shall allow users to securely log out.

3.1.2 Course Management (Instructor)

- **FR-07:** The system shall allow instructors to create courses with title, description, category, price, and thumbnail.
- **FR-08:** Newly created courses shall be saved as drafts by default.
- **FR-09:** Only published courses shall be visible to students.
- **FR-10:** Instructors shall be able to add, edit, and delete lessons.
- **FR-11:** Lessons may include video URLs, document URLs, or both.
- **FR-12:** The system shall allow instructors to delete courses with confirmation.

3.1.3 Student Learning Experience

- **FR-13:** The system shall display only published courses to students.
- **FR-14:** Students shall be able to search courses by title and category.
- **FR-15:** Free courses shall enroll students immediately.
- **FR-16:** Paid courses shall trigger the Chapa payment process.
- **FR-17:** Students shall access enrolled courses via a “My Learning” dashboard.
- **FR-18:** The system shall track lesson completion and update progress.

3.1.4 Payment Processing

- **FR-19:** The system shall generate a unique transaction reference for each payment.

- **FR-20:** The system shall verify payment status with Chapa before granting course access.
- **FR-21:** All payment transactions shall be stored with status tracking.

3.1.5 Rating System

- **FR-22:** Enrolled students shall be able to rate courses from 1 to 5 stars.
- **FR-23:** The system shall dynamically calculate average ratings.
- **FR-24:** Students shall be able to update their ratings.

3.1.6 Profile and Settings

- **FR-25:** Users shall be able to update profile information.
- **FR-26:** Users shall be able to switch between Light, Dark, and System themes.

4.External Interface Requirements

4.1 User Interface

- Mobile-friendly UI designed using Flutter
- Responsive layouts for different screen sizes

4.2 Hardware Interface

- No specific hardware requirements beyond a smartphone

4.3 Software Interface

- Firebase Authentication API
- Firebase Firestore Database
- Firebase Storage
- Chapa Payment API

4.4 Communication Interface

- HTTPS-based communication with Firebase and Chapa servers

5.Non-Functional Requirements

5.1 Security

- Role-based Firestore access rules
- Users can only modify their own data
- Instructors can only manage their own courses

5.2 Performance

- Cached network images must be used
- Lazy loading for lists to improve performance

5.3 Reliability

- Graceful error handling for network and payment failures
- Partial offline support for navigation

5.4 Usability

- Simple, intuitive UI for all user roles
- Clear error messages and confirmations

6.Future Enhancements

- Push notifications
- Real-time chat between students and instructors
- Quiz and assessment module
- Automated certificate generation upon course completion